

FLATTRIKE · REQUIREMENTS

FlatTrike requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/flattrike		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 status from FTK-CAL-001; R15 budget scope redefined; new R18 production cost target (FTK-DDR-001)
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

FlatTrike requirements

These requirements are checked by calculation in FTK-CAL-001 against the TRL 3 parametric model. Targets are still proposals until co-design sessions with users (see FTK-PRB-001). Amish's decisions of 2026-09-25 (FTK-DDR-001) redefine what the prototype budget covers (R15) and add a production cost target (R18). Six requirements are **not met** at TRL 3: R1 with a 90 kg rider, R8 (empty mass), R10 (stability with no cargo), R13 without assist, R14 for the spine side plates and R18 (production cost). Proposed changes to those targets are open items in FTK-DDR-001, awaiting Amish.

The **design load case** is an 80 kg rider and 150 kg of cargo in the box on a trike of 67.5 kg: 297.5 kg gross, on a flat dry road unless stated.

Table 1. FlatTrike requirements and TRL 3 status (FTK-CAL-001).

ID	Requirement	Target	Verification	TRL 3 status
R1	Carry the rated load	150 kg of cargo in the box plus a rider up to 90 kg; gross mass 300 kg or less, within the EN 17860 cargo cycle class	Mass budget (FTK-CAL-001 section 3)	Not met with a 90 kg rider: 307.5 kg. Met with an 80 kg rider: 297.5 kg
R2	Frame strength	Frame survives a static proof load of twice the rated payload with no permanent set, and a fatigue load case representative of commercial use under EN 17860	Hand calculation (section 6); FEA next; proof test later	At risk: stresses 45 MPa or less and fatigue ranges below the limit, but stay buckling has a safety factor of only 2.06 and the headset carries 3.34 kN per bearing in roll
R3	No welding	Every steel part is flat 3 mm plate cut from a DXF file by laser or waterjet; no bending, welding or brazing; joints bolted with locknuts	Model review	Met: 27 plates and 72 spacer washers, all flat
R4	One standard sheet	All steel frame parts nest on one 1,250 x 2,500 x 3 mm sheet (4 x 8 ft class)	Nesting check (section 2)	Met: 2,486 mm of sheet used by bounding rectangles (14 mm margin)
R5	Standard bicycle parts	Wheels (20 in, ISO 406), hubs, brakes, headset (1 1/8 in), bottom bracket, cranks, chain (1/8 in), seatpost (27.2 mm) and saddle are standard sizes sold in regional markets	Parts availability survey with the partner	Met by design, with the head tube and bottom bracket shell as bought steel sections (decided); survey pending
R6	Ships flat	Complete kit, wheels included, packs into 0.4 m ³ or less with no piece longer than 1.0 m	Packing estimate	Met: about 0.39 m ³ ; longest piece 0.80 m
R7	Assembled with hand tools	Two people assemble the kit in 4 h or less with spanners to 17 mm, hex keys, cone spanners and a screwdriver; no jig, because the plates locate each other with tabs and slots	Work-content estimate; later timed build	At risk: about 3.6 h for 112 bolt sets
R8	Light enough to ride and push	Empty mass 55 kg or less, matching the improved rickshaw documented by ITDP	Mass budget	Not met: 67.5 kg (plates 28.6 kg, box 14.7 kg)
R9	Fits market lanes	Overall width 1.0 m or less; length 2.2 m or less; turning circle 6 m or less	Model and steering geometry	Met: 0.986 m wide, 2.11 m long, 5.67 m turning circle at 35° lock
R10	Resist tipping in turns	Lateral acceleration at the tipping point 0.30 g or more, both loaded and with the rider only	Stability calculation (section 5); later tilt-table test	Not met with no cargo: 0.24 g straight and 0.11 g at full box-steering lock. Met loaded: 0.43 g
R11	Stop and park safely	Stop from 15 km/h in 6 m or less at gross mass on a dry road; drum brakes on all three wheels; a parking brake that holds the loaded trike on a 10 % grade	Braking calculation (section 7); later field test	At risk: each drum needs about 36 to 37 N·m (rating not yet confirmed); fade after about 1.9 km of loaded 5 % descent
R12			Power calculation	Met: 7.6 km/h

ID	Requirement	Target	Verification	TRL 3 status
	Pedal on the flat	Cruise at 7 km/h or more at gross mass on a flat dirt road with 110 W at the pedals		
R13	Climb local hills	Climb a 5 % grade for 100 m at 4 km/h or more at gross mass without the rider dismounting	Power and gearing calculation	Not met without assist: 238 W needed at the pedals. Met with assist route C (6.0 km/h), which is not in the prototype
R14	Repairable locally	Any single plate or standard part replaced in 30 min or less with hand tools; open DXF files let any laser shop re-cut one plate	Bolt-count estimate; later repair trial	Not met for the spine side plates: about 78 min (42 bolts). Fork plates about 24 min
R15	Affordable prototype	Parts for the pedal-only prototype cost \$800 or less (<code>project.yaml</code> budget). The optional assist kit and any SwapCell pack are outside this budget; a SwapCell pack is priced once in the SwapCell repo	Priced BOM (<code>bom/bom.csv</code>)	Met: \$647
R16	Last in commercial service	Frame life of 5 years or more in daily outdoor commercial use, against 2 to 3 years for traditional wooden rickshaw frames (ITDP)	Corrosion and fatigue review; later field trial	Not verifiable at TRL 3: unprotected steel could lose up to 0.42 mm in 5 years; depends on coating and sealed joints
R17	Work as a stall	Lockable box of 150 L or more; box lid usable as a counter at 0.75 to 0.95 m height	Model check	Met: 184 L inside; lid at 0.77 m
R18	Competitive in production	Production cost of \$300 or less per trike at 100 units (decided by Amish, 2026-09-25)	Cost estimate (section 9); later partner quotes	Not met: about \$414, two thirds of it bought bicycle parts

■ Assumptions

- Rolling resistance coefficient about 0.015 on a mix of worn asphalt and packed dirt; drag area about 0.9 m² for an upright rider behind a cargo box; air density 1.2 kg/m³.
- The rider sustains about 110 W at the pedals for an hour; short efforts can reach 200 W or more.
- Drivetrain efficiency (chain and 3-speed hub) about 89 %.
- Plate is 3 mm hot-rolled mild steel (S235JR or ASTM A36 class), density 7,850 kg/m³.
- Plywood is 9 to 12 mm exterior grade, about 700 kg/m³.
- Volume prices (R18) assume one full sheet at \$1.0/kg, cutting at \$0.30/m and bought parts at 65 % of prototype retail; no quotes yet.
- The 300 kg gross limit follows the EN 17860 class for single and multi-track cargo cycles; the standard's detailed test loads have not yet been checked against this design.